

ELITE MATHEMATICS

Pearson Edexcel International A Level

Pure 1

Adaptive Classified

Chapter 01 • Indices and Surds

Questions

Student book • questions with writing space

- 33 pages
- Past papers up to June 2026

Indices and Surds

Your classified route

12
questions

73
marks

3
families

7
variants

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CLASSIFIED BY IDEA

Move from recognition to method, then use the working space to complete the solution.

Indices and Surds

Families and variants

F01 • Index Laws and Re-expression

V01 Algebraic Fractional and Negative Powers

V02 Re-expression Through a Named Power

V03 Single Common-base Index Equation

F02 • Power-substitution Equations

V01 Quadratic in an Exponential Power

V02 Fractional-power Equation to a Polynomial

F03 • Exact Surd Algebra

V01 Simplify, Combine and Expand Surds

V02 Rationalise a Binomial Surd Denominator

V03 Solve an Exact Linear Surd Equation

V04 Nested-surd Reciprocal Identity

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Index Laws and Re-expression

WHAT THIS FAMILY ASKS

Simplify expressions by rewriting compatible bases and applying the laws of rational indices, often to express the result in a prescribed variable or base.

VARIANT ROUTE

V01 Algebraic Fractional and Negative Powers

V02 Re-expression Through a Named Power

3 Complete Classified questions

14 total marks

2. Given that

$$a = \frac{1}{64}x^2 \quad b = \frac{16}{\sqrt{x}}$$

express each of the following in the form kx^n where k and n are simplified constants.

(a) $a^{\frac{1}{2}}$

(1)

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2. Given that

$$a = \frac{1}{64}x^2 \quad b = \frac{16}{\sqrt{x}}$$

express each of the following in the form kx^n where k and n are simplified constants.

(b) $\frac{16}{b^3}$

(1)

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2. Given that

$$a = \frac{1}{64}x^2 \quad b = \frac{16}{\sqrt{x}}$$

express each of the following in the form kx^n where k and n are simplified constants.

(c) $\left(\frac{ab}{2}\right)^{-\frac{4}{3}}$

(2)

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1. Given that

$$p = \frac{1}{16}x^4 \qquad q = \frac{40}{x^3}$$

express each of the following in the form kx^n where k and n are fully simplified constants.

(a) $p^{\frac{1}{2}}$

(1)

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1. Given that

$$p = \frac{1}{16}x^4 \qquad q = \frac{40}{x^3}$$

express each of the following in the form kx^n where k and n are fully simplified constants.

(b) $(pq)^{-1}$

(2)

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1. Given that

$$p = \frac{1}{16}x^4 \qquad q = \frac{40}{x^3}$$

express each of the following in the form kx^n where k and n are fully simplified constants.

(c) pq^2

(2)

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2. Given $y = 3^x$, express each of the following in terms of y . Write each expression in its simplest form.

(a) 3^{3x}

(1)

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2. Given $y = 3^x$, express each of the following in terms of y . Write each expression in its simplest form.

(b) $\frac{1}{3^{x-2}}$

(2)

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2. Given $y = 3^x$, express each of the following in terms of y . Write each expression in its simplest form.

(c) $\frac{81}{9^{2-3x}}$

(2)

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Power-substitution Equations

WHAT THIS FAMILY ASKS

Replace a repeated power expression by one variable, solve the resulting polynomial and then return to the original variable with the correct domain filter.

VARIANT ROUTE

V01 Quadratic in an Exponential Power

3 Complete Classified questions

21 total marks

5.

In this question you must show all stages of your working.

Solutions relying on calculator technology are not acceptable.

(a) By substituting $p = 3^x$, show that the equation

$$3 \times 9^x + 3^{x+2} = 1 + 3^{x-1}$$

can be rewritten in the form

$$9p^2 + 26p - 3 = 0$$

(3)

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$$p = 3^x$$

$$9p^2 + 26p - 3 = 0$$

(b) Hence solve

$$3 \times 9^x + 3^{x+2} = 1 + 3^{x-1}$$

(3)

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4.

In this question you must show all stages of your working.

Solutions relying on calculator technology are not acceptable.

(a) By substituting $p = 2^x$, show that the equation

$$2 \times 4^x - 2^{x+3} = 17 \times 2^{x-1} - 4$$

can be written in the form

$$4p^2 - 33p + 8 = 0$$

(3)

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$$p = 2^x$$

$$4p^2 - 33p + 8 = 0$$

(b) Hence solve

$$2 \times 4^x - 2^{x+3} = 17 \times 2^{x-1} - 4$$

(3)

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In this question you must show all stages of your working. Solutions relying entirely on calculator technology are not acceptable.

$$y = a^x \quad \text{where } a \text{ is a positive constant}$$
$$(a) \quad a^{3x+1} \tag{1}$$

$$(b) \quad \frac{5}{(3a^{1-x})^{-2}} \quad (3)$$

(ii) (a) Use the substitution $p = 9^t$ to show that the equation

$$3(3^{4t+2} + 1) = 82 \times 9^t$$

can be rewritten as

$$27p^2 - 82p + 3 = 0 \quad (2)$$

(b) Hence solve

$$3(3^{4t+2} + 1) = 82 \times 9^t \quad (3)$$

Exact Surd Algebra

WHAT THIS FAMILY ASKS

Simplify, rationalise and solve exact expressions involving square roots while preserving an exact requested form.

VARIANT ROUTE

V02 Rationalise a Binomial Surd Denominator

V03 Solve an Exact Linear Surd Equation

V04 Nested-surd Reciprocal Identity

6 Complete Classified questions

38 total marks

3.

In this question you must show all stages of your working.

Solutions relying on calculator technology are not acceptable.

- (i) Show that $\frac{\sqrt{180} - \sqrt{80}}{\sqrt{5}}$ is an integer and find its value.

(2)

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(ii) Simplify

$$\frac{4\sqrt{5} - 5}{7 - 3\sqrt{5}}$$

giving your answer in the form $a + b\sqrt{5}$ where a and b are rational numbers.

(3)

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In this question you must show all stages of your working.

(i)

$$f(x) = (x + \sqrt{2})^2 + (3x - 5\sqrt{8})^2$$

Express $f(x)$ in the form $ax^2 + bx\sqrt{2} + c$ where a, b and c are integers to be found. (3)

(ii) Solve the equation

$$\sqrt{3}(4y - 3\sqrt{3}) = 5y + \sqrt{3}$$

giving your answer in the form $p + q\sqrt{3}$ where p and q are simplified fractions to be found.

(4)

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3.

In this question you must show all stages of your working.

Solutions relying on calculator technology are not acceptable.

(a) Write $\frac{8 - \sqrt{15}}{2\sqrt{3} + \sqrt{5}}$ in the form $a\sqrt{3} + b\sqrt{5}$ where a and b are integers to be found.

(3)

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(b) Hence, or otherwise, solve

$$(x + 5\sqrt{3})\sqrt{5} = 40 - 2x\sqrt{3}$$

giving your answer in simplest form.

(3)

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2. (i) Given that $m = 2^n$, express each of the following in simplest form in terms of m .

(a) 2^{n+3}

(1)

(b) 16^{3n}

(2)

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(ii)

In this question you must show all stages of your working.

Solutions relying on calculator technology are not acceptable.

Solve the equation

$$x\sqrt{3} - 3 = x + \sqrt{3}$$

giving your answer in the form $p + q\sqrt{3}$ where p and q are integers.

(3)

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2.

In this question you must show all stages of your working.

Solutions relying on calculator technology are not acceptable.

(i) Simplify fully

$$\frac{3y^3(2x^4)^3}{4x^2y^4}$$

(3)

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(ii) Find the exact value of a such that

$$\frac{16}{\sqrt{3} + 1} = a\sqrt{27} + 4$$

Write your answer in the form $p\sqrt{3} + q$ where p and q are fully simplified rational constants.

(4)

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6.

In this question you must show all stages of your working.

Solutions relying on calculator technology are not acceptable.

(a) Expand and simplify

$$\left(r - \frac{1}{r}\right)^2$$

(2)

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6.

In this question you must show all stages of your working.

Solutions relying on calculator technology are not acceptable.

(b) Express $\frac{1}{3 + 2\sqrt{2}}$ in the form $p + q\sqrt{2}$ where p and q are integers.

(2)

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6.

In this question you must show all stages of your working.

Solutions relying on calculator technology are not acceptable.

(c) Use the results of parts (a) and (b), or otherwise, to show that

$$\sqrt{3 + 2\sqrt{2}} - \frac{1}{\sqrt{3 + 2\sqrt{2}}} = 2$$

(3)

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